

Exercise 1.8

Starting from the representation of transition amplitude in 1-dimensional quantum mechanics in terms of Heisenberg picture evolution operator, ..., derive the path integral representation.

Heisenberg picture: $K(a, b) \equiv \langle x_b, t_b | x_a, t_a \rangle = \langle x_b | e^{-iH T} | x_a \rangle \quad || \quad T = t_b - t_a$

We start by discretizing the time evolution operator

$$e^{-iHT} = e^{-iH(\frac{T}{N} + \frac{T}{N} + \frac{T}{N} + \dots)} = \prod_{i=1}^N e^{-iH\epsilon} \quad , \text{ where } \epsilon = \frac{T}{N} \text{ is small time slice}$$

$$\Rightarrow K = \langle x_b | \underbrace{e^{-iH\epsilon} \dots e^{-iH\epsilon}}_{N \text{ pieces}} | x_a \rangle \quad || \text{ We insert } 1 = \int dx_i |x_i\rangle \langle x_i| \text{ for every time interval}$$

$$\Rightarrow K = \int dx_1 dx_2 dx_3 \dots dx_{N-1} \langle x_b | e^{-iH\epsilon} | x_{N-1} \rangle \langle x_{N-1} | e^{-iH\epsilon} | x_{N-2} \rangle \dots \langle x_1 | e^{-iH\epsilon} | x_a \rangle \quad || \begin{array}{l} \text{we set} \\ x_b = x_N \\ x_a = x_0 \end{array}$$

$$K = \int dx_1 dx_2 \dots dx_{N-1} K(x_N, x_{N-1}) \dots K(x_1, x_0) \quad || \quad K(x_i, x_j) = \langle x_i | e^{-iH\epsilon} | x_j \rangle$$

Now let's look what the ^{infinitesimal} transition amplitudes $K(x_{j+1}, x_j)$ are when expanded

$$\begin{aligned} \langle x_{j+1} | e^{-iH\epsilon} | x_j \rangle &= \langle x_{j+1} | e^{-i\epsilon(\frac{p^2}{2m} + V)} | x_j \rangle \quad \left\{ \begin{array}{l} e^A e^B = e^{A+B+[A,B]/2+\dots} \\ \text{the terms } [A,B] \text{ are proportional} \\ \text{to higher order terms of } O(\epsilon^2) \\ \text{of } \epsilon \text{ and since } \epsilon \text{ is small} \\ \text{we neglect them.} \end{array} \right. \\ &= e^{-i\epsilon V} \langle x_{j+1} | e^{-i\epsilon \frac{p^2}{2m}} | x_j \rangle \quad || 1 = \int dp |p\rangle \langle p| \quad \langle x|p\rangle = \frac{e^{ipx}}{\sqrt{2\pi}} \\ &= e^{-i\epsilon V} \int dp \langle x_{j+1} | p \rangle e^{-i\epsilon \frac{p^2}{2m}} \langle p | x_j \rangle \quad \Rightarrow \langle p | x \rangle = e^{-ipx} \cdot \frac{1}{\sqrt{2\pi}} \Rightarrow e^{-i\epsilon \frac{p^2}{2m} - i\epsilon V} = e^{-i\epsilon \frac{p^2}{2m}} e^{-i\epsilon V} \\ &= e^{-i\epsilon V} \frac{1}{2\pi} \int dp e^{ip \cdot x_{j+1}} e^{-i\epsilon \frac{p^2}{2m}} e^{-ip \cdot x_j} \quad || e^{ipx_{j+1}} e^{-i\epsilon \frac{p^2}{2m}} e^{-ipx_j} = e^{-i\epsilon \frac{p^2}{2m} + ip(x_{j+1} - x_j)} \end{aligned}$$

$$= e^{-i\epsilon V} \frac{1}{2\pi} \int dp e^{-i\epsilon \frac{p^2}{2m} + ip(x_{j+1} - x_j)} = e^{-i\epsilon V} \frac{1}{2\pi} \int dp e^{-\left[\frac{i\epsilon}{2m} p^2 - i(x_{j+1} - x_j) p \right]}$$

$$= e^{-i\epsilon V} \frac{1}{2\pi} \sqrt{\frac{2m\pi}{i\epsilon}} e^{-\frac{i^2 2m}{4i\epsilon} (x_{j+1} - x_j)^2} \quad || \text{ Gaussian integral } \int dx e^{-(ax^2 + bx)} = \sqrt{\frac{\pi}{a}} e^{-\frac{b^2}{4a}}$$

$$= e^{-i\epsilon V} \sqrt{\frac{m}{2\pi i\epsilon}} e^{im(x_{j+1} - x_j)^2 / 2\epsilon} \quad || \text{ Lagrangian: } m \frac{(x_{j+1} - x_j)^2}{2\epsilon^2} - V \quad \left(\begin{array}{l} \text{from time} \\ t_j \text{ to } t_j + \epsilon \end{array} \right)$$

$$= \sqrt{\frac{m}{2\pi i\epsilon}} \exp \left[i\epsilon \left(m \frac{(x_{j+1} - x_j)^2}{2\epsilon^2} - V \right) \right] = \sqrt{\frac{m}{2\pi i\epsilon}} \exp [i\epsilon L(t_j)]$$

Now we put this expression into the full transition amplitude K

$$K(t_b, t_a) = \int dx_1 dx_2 \dots dx_{N-1} \prod_{j=0}^{N-1} \sqrt{\frac{m}{2\pi i \epsilon}} \exp \left[i \epsilon \left(\frac{(x_{j+1} - x_j)^2}{\epsilon^2} - V(x_j, t_j) \right) \right]$$

$$\prod_{j=0}^{N-1} e^x = e^{\sum_{j=0}^{N-1} x}$$

$$\Rightarrow K(t_b, t_a) = \int dx_1 dx_2 \dots dx_{N-1} \left(\frac{m}{2\pi i \epsilon} \right)^{\frac{N}{2}} \exp \left[i \sum_{j=0}^{N-1} \underbrace{\left(\frac{(x_{j+1} - x_j)^2}{\epsilon^2} m - V(x_j, t_j) \right)}_L \epsilon \right]$$

$\epsilon = \frac{T}{N} = \delta t$ is very small time interval. We infinitesimize δt by $N \rightarrow \infty$, $\delta t \rightarrow 0$

So the sum becomes integral:

$$K(t_b, t_a) = \left(\frac{m}{2\pi i \epsilon} \right)^{N/2} \int dx_1 dx_2 \dots dx_{N-1} \exp \left[i \int_{t_a}^{t_b} dt L(t) \right] \quad \left\| \begin{array}{l} \text{Now we write} \\ \int D x(t) \equiv \left(\frac{m}{2\pi i \epsilon} \right)^{N/2} \int dx_1 dx_2 \dots dx_{N-1} \end{array} \right.$$

$$= \int D x(t) \exp \left[i \int_{t_a}^{t_b} L(t) \right]$$

$$\underline{K(t_b, t_a) = \int D x(t) \exp [i S[x(t)]]} \quad \square, \quad \text{here } S[x(t)] \text{ is the action } \int_{t_a}^{t_b} L(t)$$

We have derived the path integral representation. $\square$

Exercise 1.9

Derive the 4-point function in Klein-Gordon theory starting from the functional integral representation of the generating function

$$Z_{KG}[J] = \exp \left[\frac{i}{2} \int d^4x d^4y J(x) i \Delta_F(x-y) J(y) \right]$$

The four-point function is

$$+ \left(\prod_{i=1}^4 \frac{\delta}{\delta J(x_i)} \right) Z[J] \Big|_{J=0}$$

There might've been wrong sign in the slides

Here and after !!
we write
 $Z_{KG}[J] \equiv Z[J]$

$$\Rightarrow F = + \frac{\delta}{\delta J(x_4)} \frac{\delta}{\delta J(x_3)} \frac{\delta}{\delta J(x_2)} \frac{\delta}{\delta J(x_1)} \exp \left[\frac{i}{2} \int d^4x d^4y J(x) i \Delta_F(x-y) J(y) \right] \Big|_{J=0}$$

$$= + \left(\prod_{i=2}^4 \frac{\delta}{\delta J(x_i)} \right) \frac{i}{2} \left[\int d^4x d^4y \delta(x_1-x) i \Delta_F(x-y) J(y) + \int d^4x J(x) i \Delta_F(x-x_1) \right] Z[J] \Big|_{J=0}$$

Here the integration over y already done

Here we have used the basic axioms for functional derivatives

$$\frac{\delta}{\delta J(x)} J(y) = \delta^{(4)}(x-y) \quad \text{and} \quad \frac{\delta}{\delta J(x)} \int d^4y J(y) \phi(y) = \phi(x)$$

$$\Rightarrow F = + \left(\prod_{i=3}^4 \frac{\delta}{\delta J(x_i)} \right) \frac{\delta}{\delta J(x_2)} \underbrace{i \left[\int d^4 y \Delta_F(x_1-y) J(y) + \int d^4 x J(x) \Delta_F(x-x_1) \right]}_{= C(x_1)} Z[J] \Big|_{J=0}$$

$$= + \frac{i}{2} \left(\prod_{i=3}^4 \frac{\delta}{\delta J(x_i)} \right) \left[\underbrace{i \Delta_F(x_1-x_2) + i \Delta_F(x_2-x_1)}_{= 2i \Delta_F(x_1-x_2)} Z[J] + C(x_1) \frac{\delta}{\delta J(x_2)} Z[J] \right] \Big|_{J=0}$$

$$= + \frac{\delta}{\delta J(x_4)} \left[i^2 \Delta_F(x_1-x_2) C(x_3) Z[J] + \frac{\delta}{\delta J(x_3)} [C(x_1) C(x_2) Z[J]] \right] \Big|_{J=0}$$

$$= + \Delta_F(x_1-x_2) \Delta_F(x_3-x_4) Z[J] + \Delta_F(x_1-x_2) C(x_3) C(x_4) Z[J]$$

↑ Becomes 0 when $J=0$

$$+ \frac{\delta}{\delta J(x_4)} \left(- \Delta_F(x_1-x_3) C(x_2) Z[J] + C(x_1) \Delta_F(x_2-x_3) Z[J] + \underbrace{C(x_1) C(x_2) C(x_3) Z[J]}_{\substack{\uparrow \text{ This also} \\ \text{becomes 0 when } J=0}} \right) \Big|_{J=0}$$

$$= + \Delta_F(x_1-x_2) \Delta_F(x_3-x_4) Z[J] + \Delta_F(x_1-x_3) \Delta_F(x_2-x_4) Z[J] + \Delta_F(x_1-x_4) \Delta_F(x_2-x_3) Z[J] \Big|_{J=0}$$

$$Z[J] \Big|_{J=0} = Z_{\text{cl}}[J] \Big|_{J=0} = 1$$

$$\Rightarrow F = \left(\prod_{k=1}^4 \frac{-i\delta}{\delta J(x_k)} \right) Z[J] \Big|_{J=0} = \frac{\Delta_F(x_1-x_2) \Delta_F(x_3-x_4) + \Delta_F(x_1-x_3) \Delta_F(x_2-x_4) + \Delta_F(x_1-x_4) \Delta_F(x_2-x_3)}{\square}$$

The four-point function is :

$$\Delta_F(x_1-x_2) \Delta_F(x_3-x_4) + \Delta_F(x_1-x_3) \Delta_F(x_2-x_4) + \Delta_F(x_1-x_4) \Delta_F(x_2-x_3)$$

Exercise 1.10

Consider the generating function of the Klein-Gordon theory. Show that the normalization factor of $Z[J]$ can be expressed as

$$N = \int D\phi e^{i \int d^4 x \mathcal{L}_{KG}} = \text{const.} [\text{Det}(\partial^2 + m^2)]^{-1/2}$$

$$\mathcal{L}_{KG} = \frac{1}{2} (\partial_\mu \phi \partial^\mu \phi - m^2 \phi^2) = \frac{1}{2} (\partial \phi)^2 - \frac{1}{2} m \phi^2$$

$$\mathcal{L}_{KG} = -\frac{1}{2} [\phi (\partial^2 + m^2) \phi - \partial_\mu (\phi \partial^\mu \phi)]$$

↑ Gives a surface term

$$\begin{aligned} \partial_\mu (\phi \partial^\mu \phi) &= \phi \partial_\mu \partial^\mu \phi + (\partial_\mu \phi) (\partial^\mu \phi) \\ &= \phi \partial^2 \phi + (\partial \phi)^2 \\ \Rightarrow (\partial \phi)^2 &= \partial_\mu (\phi \partial^\mu \phi) - \phi \partial^2 \phi \end{aligned}$$

$$\Rightarrow N = \int D\phi e^{-i \frac{1}{2} \int d^4 x \phi (\partial^2 + m^2) \phi + (\text{Surface term})}$$

Theorem 1: For $n \times n$ matrix the product of it's n eigenvalues is same as the determinant.

Theorem 2: For $n \times n$ matrix it's Trace is a sum of it's n eigenvalues

For integral of the form $\left(\prod_k d\xi_k \right) \exp[-\xi_i B_{ij} \xi_j]$

with B being symmetric matrix with eigenvalues b_i . We write $\xi_i = O_{ij} x_j$

where O is orthogonal matrix of eigenvectors that diagonalizes B .

Doing change of variables from ξ_i to x_i we get

K_{ij} is the diagonalized Matrix x

$$\left(\prod_k d\xi_k \right) \exp(-\xi_i B_{ij} \xi_j) = \left(\prod_k dx_k \right) \exp\left[-x_i \overset{\vee}{K}_{ij} x_j\right]$$

$$= \left(\prod_k dx_k \right) \exp\left[-\text{Tr}(x K x)\right] \quad \parallel \text{ Use Theorem 2 : } \text{Tr}(x K x) = \sum_i b_i x_i^2$$

$$= \left(\prod_k dx_k \right) \exp\left[-\sum_i b_i x_i^2\right] = \prod_i \left(\int dx_i \exp(-b_i x_i^2) \right) \parallel \text{Gauss integral}$$

$$= \prod_i \sqrt{\frac{\pi}{b_i}} \quad \parallel \text{Now we use theorem 1}$$

$$= \text{const} \cdot [\det B]^{-1/2}$$

Now we go back to our normalization factor

$$N = \int D\phi e^{-\frac{i}{2} \int d^4x \phi (\partial^2 + m) \phi + \text{surface term}}$$

Surface term vanishes (assumption)

$$\text{We discretize the integral to get } N = \int D\phi e^{-\frac{i}{2} \sum_i \phi_i (\partial^2 + m)_{ii} \phi_i} \parallel \int D\phi = \prod_i \int d\phi_i$$

$$N = \prod_i \left(\int d\phi_i \exp\left(-\frac{i}{2} \phi_i (\partial^2 + m)_{ii} \phi_i\right) \right)$$

Now we use the result that we derived for matrix B to get

$$N = \text{const} \cdot [\det(\partial^2 + m)]^{-1/2}$$

the $(\partial^2 + m)_{ii}$ operator acts on field $\phi_i = \phi(x_i)$ we can think of the determinant being of matrix of type

$$\begin{pmatrix} (\partial^2 + m)_{11} & \dots & 0 & \dots & 0 \\ 0 & \dots & (\partial^2 + m)_{22} & \dots & 0 \\ \vdots & \dots & 0 & \dots & (\partial^2 + m)_{ii} \end{pmatrix}$$

So the determinant would depend on how many i 's there are and

I have no idea how to evaluate this when $i \rightarrow \infty$.